

Day 3 Assignment

30-12-25

CREATE DATABASE

```
create database airlineManagementDB;  
use airlineManagementDB;
```

CREATE TABLES

CREATE air_flight table

```
create table air_flight(  
flight_id varchar(10) primary key,  
airline_id varchar(10) not null,  
airline_name varchar(30) not null,  
from_location varchar(20) not null,  
to_location varchar(20) not null,  
departure_time time not null,  
arrival_time time not null,  
duration time not null,  
total_seats int not null  
);
```

CREATE air_flight_details table

```
create table air_flight_details(  
id int identity(1,1) primary key,
```

```
flight_id varchar(10) not null ,  
flight_departure_date date unique not null,  
price decimal(8,2),  
available_seats int,  
foreign key(flight_id) references air_flight(flight_id)  
)
```

CREATE air_passenger_profile table

```
create table air_passenger_profile(  
profile_id varchar(10) primary key,  
password varchar(100),  
first_name varchar(10),  
last_name varchar(10),  
address varchar(100),  
mobile_number varchar(15),  
email_id varchar(30)  
)
```

CREATE air_credit_card_details table

```
create table air_credit_card_details(  
id int primary key identity(1,1),  
profile_id varchar(10),  
card_number varchar(16),
```

```
card_type varchar(10),  
expiration_month int,  
expiration_year int,  
foreign key (profile_id) references air_passenger_profile(profile_id)  
)
```

CREATE air_ticket_info table

```
create table air_ticket_info(  
ticket_id varchar(10) primary key,  
profile_id varchar(10) not null,  
flight_id varchar(10),  
flight_departure_date date,  
status varchar(10),  
foreign key(profile_id) references air_passenger_profile(profile_id),  
foreign key(flight_id) references air_flight(flight_id),  
foreign key (flight_departure_date) references  
air_flight_details(flight_departure_date)  
)
```

INSERTION COMMANDS

Insert values in air_flight table

insert into air_flight values

```
('F101','A01','ABC Airlines','Chennai','Hyderabad','06:00','07:30','01:30',180),  
( 'F102','A01','ABC Airlines','Hyderabad','Chennai','09:00','10:30','01:30',180),  
( 'F103','A01','ABC Airlines','Chennai','Bangalore','11:00','12:00','01:00',150),  
( 'F104','A01','ABC Airlines','Delhi','Mumbai','14:00','16:00','02:00',200),  
( 'F105','A01','ABC Airlines','Mumbai','Delhi','18:00','20:00','02:00',200),  
( 'F106','A01','ABC Airlines','Chennai','Hyderabad','21:00','22:30','01:30',180),  
( 'F107','A01','ABC Airlines','Bangalore','Chennai','08:00','09:00','01:00',150),  
( 'F108','A01','ABC Airlines','Hyderabad','Bangalore','13:00','14:15','01:15',160),  
( 'F109','A01','ABC Airlines','Delhi','Chennai','07:00','09:30','02:30',220),  
( 'F110','A01','ABC Airlines','Mumbai','Hyderabad','16:00','18:00','02:00',190);
```

Insert values in air_flight_details table

insert into air_flight_details (flight_id, flight_departure_date, price,
available_seats) values

```
('F101','2025-03-10',4500,50),  
( 'F101','2025-04-12',4800,40),  
( 'F101','2025-05-15',5000,30),  
( 'F102','2025-04-05',4600,60),  
( 'F102','2025-05-18',4900,55),
```

('F103','2025-04-20',3500,70),
('F103','2025-06-10',3700,65),
('F104','2025-04-25',6000,80),
('F105','2025-05-05',6100,75),
('F106','2025-04-18',4700,45),
('F107','2025-03-22',3300,60),
('F108','2025-04-28',3900,50);

Insert values in air_flight_details table

insert into air_flight_details (flight_id, flight_departure_date, price, available_seats) values

('F101','2024-03-05',4400,60),
('F101','2024-03-18',4600,55),
('F101','2024-04-02',4700,50),
('F101','2024-04-15',4800,45),
('F101','2024-05-10',5000,40),
('F102','2024-04-05',4500,65),
('F102','2024-04-22',4700,60),
('F102','2024-05-08',4900,55),
('F103','2024-03-12',3300,70),
('F103','2024-04-10',3500,68),
('F103','2024-04-25',3600,65),
('F103','2024-06-05',3800,60),

```
('F104','2024-02-20',5800,90),  
( 'F104','2024-04-12',6000,85),  
( 'F104','2024-04-28',6200,80),  
( 'F105','2024-03-15',5900,88),  
( 'F105','2024-05-18',6100,82),  
( 'F106','2024-04-08',4600,58),  
( 'F106','2024-04-21',4800,54),  
( 'F106','2024-05-25',5000,50),  
( 'F107','2024-03-22',3200,72),  
( 'F107','2024-04-18',3400,70),  
( 'F108','2024-04-03',3800,66),  
( 'F108','2024-06-14',4000,62);
```

Insert values in air_passenger_profile table

insert into air_passenger_profile values

```
('P01','passRavi','Ravi','Kumar','Chennai','9876543210','ravi@gmail.com'),  
( 'P02','passAnita','Anita','Sharma','Hyderabad','9876543222','anita@gmail.com'),  
( 'P03','passSuresh','Suresh','Reddy','Bangalore','9876543333','suresh@gmail.com')  
,  
( 'P04','passMeena','Meena','Iyer','Chennai','9876544444','meena@gmail.com'),  
( 'P05','passArjun','Arjun','Verma','Delhi','9876545555','arjun@gmail.com'),  
( 'P06','passPriya','Priya','Singh','Mumbai','9876546666','priya@gmail.com'),
```

```
('P07','passKiran','Kiran','Rao','Hyderabad','9876547777','kiran@gmail.com'),  
('P08','passNeha','Neha','Gupta','Delhi','9876548888','neha@gmail.com'),  
('P09','passVikram','Vikram','Patel','Ahmedabad','9876549999','vikram@gmail.co  
m'),  
('P10','passLakshmi','Lakshmi','Nair','Chennai','9876500000','lakshmi@gmail.com')  
;
```

Insert values in air_credit_card_details table

insert into air_credit_card_details (profile_id, card_number, card_type,
expiration_month, expiration_year) values

```
('P01','4111111111111111','VISA',12,2026),  
('P02','4222222222222222','MASTER',11,2027),  
('P03','4333333333333333','VISA',10,2025),  
('P04','4444444444444444','RUPAY',9,2028),  
('P05','4555555555555555','VISA',8,2026),  
('P06','4666666666666666','MASTER',7,2027),  
('P07','4777777777777777','VISA',6,2025),  
('P08','4888888888888888','RUPAY',5,2028),  
('P09','4999999999999999','VISA',4,2026),  
('P10','4000000000000000','MASTER',3,2027);
```

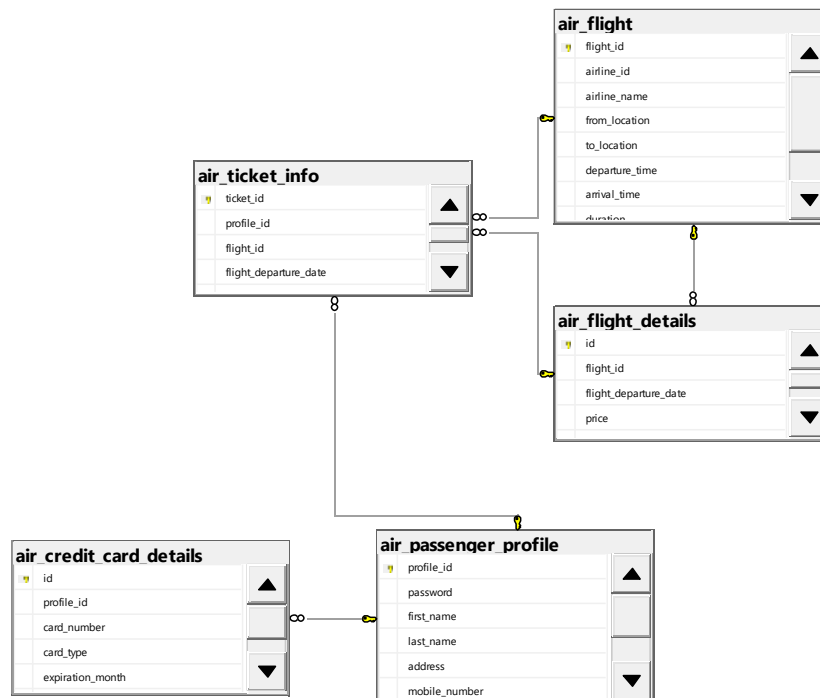
Insert values in air_ticket_info table

insert into air_ticket_info values

```
('T001','P01','F101','2025-03-10','Booked'),
```

('T002','P01','F101','2025-04-12','Booked'),
 ('T003','P01','F101','2025-04-12','Booked'),
 ('T004','P02','F101','2025-04-12','Booked'),
 ('T005','P02','F102','2025-04-05','Booked'),
 ('T006','P03','F103','2025-04-20','Booked'),
 ('T007','P04','F101','2025-05-15','Booked'),
 ('T008','P04','F101','2025-05-15','Booked'),
 ('T009','P04','F101','2025-05-15','Booked'),
 ('T010','P04','F102','2025-05-18','Booked'),
 ('T011','P05','F106','2025-04-18','Booked'),
 ('T012','P06','F108','2025-04-28','Booked');

Schema Diagram



Queries

1. Write a query to display the average monthly ticket cost for each flight in ABC Airlines. The query should display the Flight_Id, From_location, To_Location, Month Name as "Month_Name" and average price as "Average_Price". Display the records sorted in ascending order based on flight id and then by Month Name.

Query:

```
select af.flight_id,from_location,to_location,datetime(month,
flight_departure_date) as month_name,avg(price) as Average_Price

from air_flight af join air_flight_details afd on afd.flight_id=af.flight_id where
af.airline_name='ABC Airlines'

group by af.flight_id,from_location,to_location,datetime(month,
flight_departure_date),month(flight_departure_date)

order by af.flight_id,month(flight_departure_date);
```

Output:

	flight_id	from_location	to_location	month_name	Average_Price
1	F101	Chennai	Hyderabad	March	4500.000000
2	F101	Chennai	Hyderabad	April	4766.666666
3	F101	Chennai	Hyderabad	May	5000.000000
4	F102	Hyderabad	Chennai	April	4600.000000
5	F102	Hyderabad	Chennai	May	4900.000000
6	F103	Chennai	Bangalore	March	3300.000000
7	F103	Chennai	Bangalore	April	3533.333333
8	F103	Chennai	Bangalore	June	3750.000000
9	F104	Delhi	Mumbai	February	5800.000000
10	F104	Delhi	Mumbai	April	6066.666666
11	F105	Mumbai	Delhi	March	5900.000000
12	F105	Mumbai	Delhi	May	6100.000000
13	F106	Chennai	Hyderabad	April	4700.000000
14	F106	Chennai	Hyderabad	May	5000.000000
15	F107	Bangalore	Chennai	March	3250.000000
16	F107	Bangalore	Chennai	April	3400.000000
17	F108	Hyderabad	Bangalore	April	3850.000000
18	F108	Hyderabad	Bangalore	June	4000.000000

2. Write a query to display the customer(s) who has/have booked least number of tickets in ABC Airlines. The Query should display profile_id, customer's

first_name, Address and Number of tickets booked as “No_of_Tickets”. Display the records sorted in ascending order based on customer's first name.

Query:

```
select app.profile_id,first_name,address,count(*) as No_of_Tickets
from air_passenger_profile app join air_ticket_info ati on
app.profile_id=ati.profile_id join air_flight af on af.flight_id=ati.flight_id
where af.airline_name='ABC Airlines'

group by app.profile_id,first_name,address

having count(*)=(select min(count1) as mini from (select count(*) as count1 from
air_passenger_profile app join air_ticket_info ati on app.profile_id=ati.profile_id
group by app.profile_id,first_name,address) as ini)

order by first_name;
```

Output:

	profile_id	first_name	address	No_of_Tickets
1	P05	Arjun	Delhi	1
2	P06	Priya	Mumbai	1
3	P03	Suresh	Bangalore	1

3. Write a query to display the number of flight services between locations in a month. The Query should display From_Location, To_Location, Month as “Month_Name” and number of flight services as “No_of_Services”. Hint: The Number of Services can be calculated from the number of scheduled departure dates of a flight. The records should be displayed in ascending order based on From_Location and then by To_Location and then by month name.

Query:

```
select from_location,to_location,datetime(month,flight_departure_date) as
Month_Name,count(afd.flight_departure_date) as 'No.of Services'

from air_flight af join air_flight_details afd on af.flight_id=afd.flight_id
```

group by from_location,to_location,datetime(month,flight_departure_date)
order by from_location,to_location,datetime(month,flight_departure_date);

Output:

	from_location	to_location	Month_Name	No_of_Services
1	Bangalore	Chennai	April	1
2	Bangalore	Chennai	March	2
3	Chennai	Bangalore	April	3
4	Chennai	Bangalore	June	2
5	Chennai	Bangalore	March	1
6	Chennai	Hyderabad	April	6
7	Chennai	Hyderabad	March	3
8	Chennai	Hyderabad	May	3
9	Delhi	Mumbai	April	3
10	Delhi	Mumbai	February	1
11	Hyderabad	Bangalore	April	2
12	Hyderabad	Bangalore	June	1
13	Hyderabad	Chennai	April	3
14	Hyderabad	Chennai	May	2
15	Mumbai	Delhi	March	1
16	Mumbai	Delhi	May	2

4. Write a query to display the customer(s) who has/have booked maximum number of tickets in ABC Airlines. The Query should display profile_id, customer's first_name, Address and Number of tickets booked as "No_of_Tickets". Display the records in ascending order based on customer's first name.

Query:

```
select ati.profile_id,app.first_name,app.address,count(ati.ticket_id) as  
'No_of_Tickets' from  
air_passenger_profile app join air_ticket_info ati on app.profile_id=ati.profile_id  
group by ati.profile_id,app.first_name,app.address
```

having count(ati.ticket_id)=(select max(cnt) as maxi from (select
count(ati.ticket_id) as cnt from air_ticket_info ati group by ati.profile_id) as ii)
order by app.first_name;

Output:

	profile_id	first_name	address	No_of_Tickets
1	P04	Meena	Chennai	4

5. Write a query to display the number of tickets booked from Chennai to Hyderabad. The Query should display passenger profile_id,first_name,last_name, Flight_Id , Departure_Date and number of tickets booked as “No_of_Tickets”. Display the records sorted in ascending order based on profile id and then by flight id and then by departure date.

Query:

```
select
ati.profile_id,first_name,last_name,ati.flight_id,flight_departure_date,count(ticket
_id) as 'No_of_Tickets'

from air_passenger_profile app join air_ticket_info ati on
app.profile_id=ati.profile_id

join air_flight af on ati.flight_id=af.flight_id

where af.from_location='Chennai' and af.to_location='Hyderabad'

group by
from_location,to_location,ati.profile_id,first_name,last_name,ati.flight_id,flight_d
eparture_date

order by ati.profile_id,ati.flight_id,flight_departure_date;
```

Output:

	profile_id	first_name	last_name	flight_id	flight_departure_date	No_of_Tickets
1	P01	Ravi	Kumar	F101	2025-03-10	1
2	P01	Ravi	Kumar	F101	2025-04-12	2
3	P02	Anita	Sharma	F101	2025-04-12	1
4	P04	Meena	Iyer	F101	2025-05-15	3
5	P05	Arjun	Verma	F106	2025-04-18	1

6. Write a query to display flight id,from location, to location and ticket price of flights whose departure is in the month of april.

Query:

```
select afd.flight_id,from_location,to_location,price
from air_flight_details afd join air_flight af on afd.flight_id=af.flight_id
where datename(month,afd.flight_departure_date)='April';
```

Output:

	flight_id	from_location	to_location	price
1	F101	Chennai	Hyderabad	4800.00
2	F102	Hyderabad	Chennai	4600.00
3	F103	Chennai	Bangalore	3500.00
4	F104	Delhi	Mumbai	6000.00
5	F106	Chennai	Hyderabad	4700.00
6	F108	Hyderabad	Bangalore	3900.00
7	F101	Chennai	Hyderabad	4700.00
8	F101	Chennai	Hyderabad	4800.00
9	F102	Hyderabad	Chennai	4500.00
10	F102	Hyderabad	Chennai	4700.00
11	F103	Chennai	Bangalore	3500.00
12	F103	Chennai	Bangalore	3600.00
13	F104	Delhi	Mumbai	6000.00
14	F104	Delhi	Mumbai	6200.00
15	F106	Chennai	Hyderabad	4600.00
16	F106	Chennai	Hyderabad	4800.00
17	F107	Bangalore	Chennai	3400.00
18	F108	Hyderabad	Bangalore	3800.00

7. Write a query to display the average cost of the tickets in each flight on all scheduled dates. The query should display flight_id, from_location, to_location and Average price as "Price". Display the records sorted in ascending order based on flight id and then by from_location and then by to_location.

Query:

```
select af.flight_id,from_location,to_location,avg(price) as price
from air_flight_details afd join air_flight af on af.flight_id=afd.flight_id
group by af.flight_id,from_location,to_location
order by af.flight_id,from_location,to_location;
```

Output:

	flight_id	from_location	to_location	price
1	F101	Chennai	Hyderabad	4725.000000
2	F102	Hyderabad	Chennai	4720.000000
3	F103	Chennai	Bangalore	3566.666666
4	F104	Delhi	Mumbai	6000.000000
5	F105	Mumbai	Delhi	6033.333333
6	F106	Chennai	Hyderabad	4775.000000
7	F107	Bangalore	Chennai	3300.000000
8	F108	Hyderabad	Bangalore	3900.000000

8. Write a query to display the customers who have booked tickets from Chennai to Hyderabad. The query should display profile_id, customer_name (combine first_name & last_name with comma in b/w), address of the customer. Give an alias to the name as customer_name. Hint: Query should fetch unique customers irrespective of multiple tickets booked. Display the records sorted in ascending order based on profile id.

Query:

```
select distinct ati.profile_id,concat(first_name,',',last_name) as 'customer_name'
,address
```

```

from air_passenger_profile app join air_ticket_info ati on
app.profile_id=ati.profile_id

join air_flight af on ati.flight_id=af.flight_id

where from_location='Chennai' and to_location='Hyderabad'

order by profile_id;

```

Output:

	profile_id	customer_name	address
1	P01	Ravi,Kumar	Chennai
2	P02	Anita,Sharma	Hyderabad
3	P04	Meena,Iyer	Chennai
4	P05	Arjun,Verma	Delhi

9. Write a query to display profile id of the passenger(s) who has/have booked maximum number of tickets. In case of multiple records, display the records sorted in ascending order based on profile id.

Query:

```

select ati.profile_id

from air_passenger_profile app join air_ticket_info ati on
app.profile_id=ati.profile_id

group by ati.profile_id

having count(ticket_id)=(select max(cnt) from (select count(ticket_id) as cnt from
air_ticket_info group by profile_id) as iii)

order by profile_id;

```

Output:

	profile_id
1	P04

10. Write a query to display the total number of tickets as “No_of_Tickets” booked in each flight in ABC Airlines. The Query should display the flight_id, from_location, to_location and the number of tickets. Display only the flights in which atleast 1 ticket is booked. Display the records sorted in ascending order based on flight id.

Query:

```
select ati.flight_id,from_location,to_location,count(ticket_id) as 'No_of_Tickets'
from air_ticket_info ati join air_flight af on ati.flight_id=af.flight_id
where af.airline_name='ABC Airlines'
group by ati.flight_id,from_location,to_location;
```

Output:

Results of Messages				
	flight_id	from_location	to_location	No_of_Tickets
1	F101	Chennai	Hyderabad	7
2	F102	Hyderabad	Chennai	2
3	F103	Chennai	Bangalore	1
4	F106	Chennai	Hyderabad	1
5	F108	Hyderabad	Bangalore	1